

Securing Docker: Best Practices

Here are some great tips to help you secure your Docker environment, which is essential to ensure smooth operations.



Containers are being increasingly utilised by organisations to manage their application deployment lifecycle. This article discusses the importance of securing the Docker environment, a crucial tool for managing the entire container management ecosystem. It highlights the need for a secure environment to ensure the smooth operation of Docker.

Here are a few best practices that can be followed to enable this environment.

Docker daemon

Sometimes, we need to access the Docker daemon from a remote machine for this. Usually, we need to enable the tcp socket. Beware that the default setup provides an unencrypted and unauthenticated direct access to the Docker daemon, and should be secured either using the built-in HTTPS encrypted socket, or by putting a secure

web proxy in front of it. You can listen on port 2375 on all network interfaces with `-H tcp://0.0.0.0:2375`, or on a particular network interface using its IP address: `-H tcp://192.168.59.103:2375`. It is conventional to use port 2375 for unencrypted, and port 2376 for encrypted communication with the daemon.

Non-root user

Avoid using root user to run containers. Instead, we can use a normal (non-root) user which can be added to the Docker group, as assigning sudo permissions for non-root users can be dangerous. It's highly recommended to use root-less mode for dealing with containers using Docker. Use the following command for this.

```
sudo groupadd docker
sudo usermod -aG docker $USER
```